

App Dev Project Report

1. Student Details

Name: Om Shiv Verma

Roll Number: 24f3005047

Email: 24f3005047@ds.study.iitm.ac.in

About Me: I am a stand-alone student in BS Degree in Data Science and Application program by IIT Madras with a deep interest in web application development and data-driven technologies. I enjoy building applications that combine learning and user experience. As it also helps me learn better and present my skills.

2. Project Details

Project Title: Trekking Management Application

Problem Statement:

To build a **Trekking Management** web application that allows **Admin, Trek Staff, and Users (Trekksers)** to interact with the system based on their roles and access the routes that they only should according to their role

Approach:

The app was built using Flask as the backend framework with a modular structure. The admin can add treks, edit treks, view staff members, view trekkers, mark staff and trekkers as blacklisted and approved, search for staff and trekkers using their user_id or name, search for treks using trek_id or its name and also view the bookings done by the trekkers. Staff members can view treks that are assigned to them, edit trek slots and status of the assigned treks, view the participants that have booked for the trek he/she has been assigned to, staff member can also update his profile- name, contact, email. Trekkers can view the available(upcoming) treks, and book the treks from the upcoming treks by clicking on the book button, view the treks for which they have done the booking, view the treks that they have completed until present date, they can also see the treks according to the difficulty and location of the treks.

3. AI/LLM Declaration

I used **ChatGPT (GPT-5)** to assist in writing SQLAlchemy model definitions, creating bootstrap, templates html pages and for debugging of the code purpose.

The extent of AI/LLM usage is around **10–15%**, limited to **code suggestions and documentation formatting**.

All final implementation logic, debugging, and integration were done manually by me.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
CSS	To add styling and background image in project
Flask-Login	User authentication and session management
Date-Time	For managing date and time in the project
SQLite	Lightweight local database for storing user data

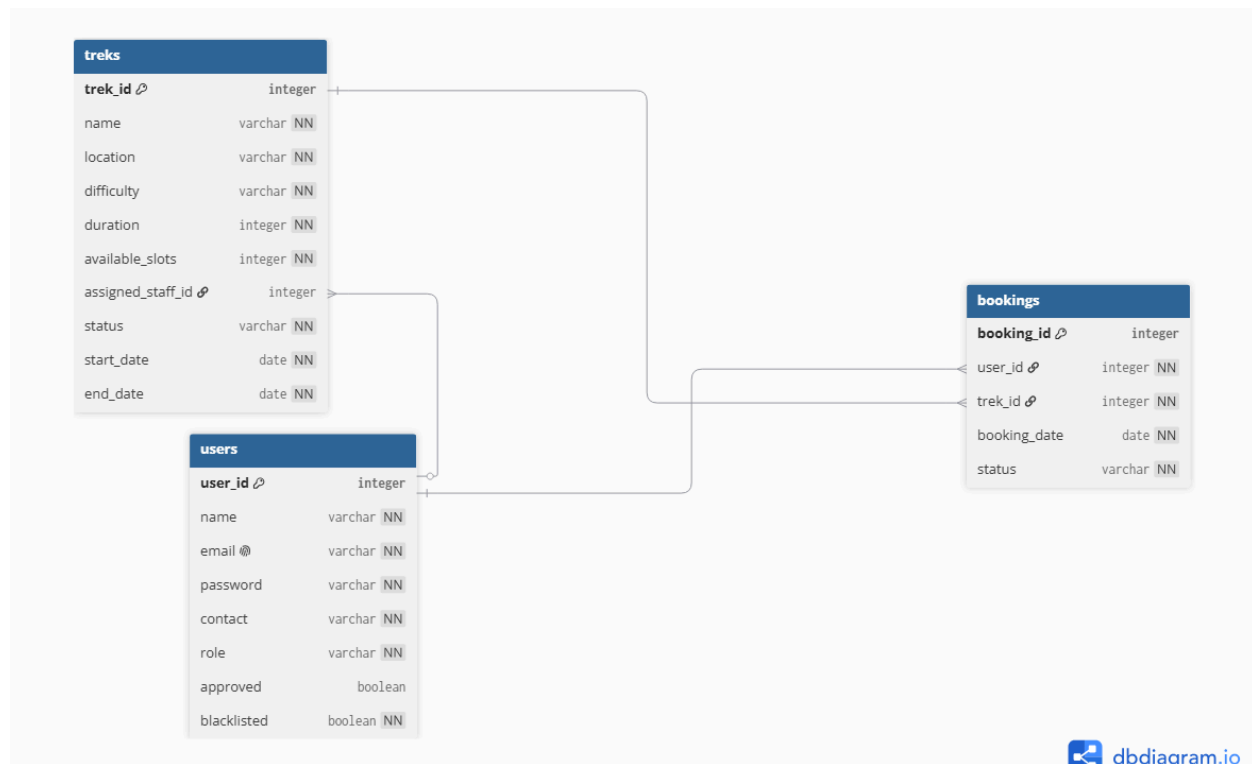
5. Database Schema / ER Diagram

Tables:

1. **User** — stores admin.staff member,trekker profile details (user id, name, email, password, contact, role, approve status, blacklisted status)
2. **Trek** — stores all the details about treks (trek id, name, location, difficulty, duration, available slots, assigned staff id, status, start date, end date)
3. **Bookings** — stores all the details about the bookings done by the trekkers (booking id, user id, trek id, booking date, booking status)

Relationships:

- One-to-Many → **Treks** → **bookings**
- One-to-Many → **users** → **treks**
- One-to-Many → **users** → **bookings**



6. Architecture and features of my project

Architecture Overview:

- app.py - Main flask application file containing routes and authentication.
- models.py- database models using SQLAlchemy
- config.py- contains the configuration
- /templates- contains all the html files and sub-directories, html using Jinja2
- /static- contains css and images folder which has css styling file and background image used in the project.

Features Implemented:

- Admin dashboard which shows the total number of trekkers, staff, bookings, treks. And has all the other pages link.
 - The pages link on admin dashboard redirects admin to pages to the following- add trek, view all treks, edit trek, view staff, view trekkers, search staff/ trekkers using their id or name and search treks using their id or name.
 - Staff dashboard shows them total number of treks assigned to them and trekkers in all the treks who made booking in the trek which was assigned to him. It also has links to updating the profile page and view the treks assigned to him/her.
 - The links from staff dashboard redirects them to the pages according to the link they press on using which they can edit their profile-name, contact, email and using which they can also view the treks assigned to them and on this page they can edit the numbers of slots or the status of a particular trek they click on and they can also view the participants in a trek they were assigned.
 - Trekker dashboard has the features same as other dashboard- links for the other pages to redirect them to the other pages, they also have a search option on the dashboard using which they can search for a trek using difficulty and location of the trek they want to book for.
 - The link of the pages from trekker dashboard redirects them to pages where they can view the upcoming treks, view the treks they have booked for, view the treks that they have completed. They can also book the treks using the upcoming treks page, delete any of their bookings using the booked treks page.
-

7. Video Presentation

Drive Link: Report video

[Video report link](#)